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A Case-Density-Driven Closed-Loop Intelligent Strategy for Air-Ground-Human Collaborative Disinfection: An End-to-End Framework from Case Perception to Task Scheduling

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Abstract: Traditional public health disinfection tasks relying on fixed-area coverage often suffer from resource waste, delayed intervention, and low response efficiency. This study proposes a case-density-driven closed-loop intelligent strategy for air-ground-human collaborative disinfection, establishing an end-to-end framework from case perception to task scheduling. Firstly, a spatiotemporal risk field is constructed based on reported case data and population mobility information, and high-risk areas are adaptively identified and prioritized through dynamic evaluation. Secondly, for coordinated execution by unmanned aerial vehicles (UAVs), ground vehicles, and personnel, a multi-objective coupled optimization model is designed, targeting coverage efficiency, suppression timeliness, path conflicts, and resource cost to generate executable collaborative schedules. Furthermore, a closed-loop execution mechanism is developed, enabling real-time rolling re-planning and adaptive strategy correction in response to task feedback, unexpected disturbances (area lockdown, equipment failure, chemical shortage), and risk field updates. Experimental results demonstrate that the proposed closed-loop approach significantly improves coverage, suppression time, and resource utilization compared with traditional static scheduling and single-entity planning methods across multiple scenarios, and exhibits robustness against environmental uncertainties and resource disturbances. This framework provides a feasible theoretical and methodological foundation for intelligent, precise, and resilient public health disinfection operations.

Keywords: Case density; Air-ground-human collaboration; Closed-loop scheduling; Intelligent disinfection; Multi-objective optimization

1 Introduction

The high frequency and increasing complexity of public health emergencies have imposed more stringent requirements on the response speed, resource allocation, and operational efficiency of urban epidemic prevention systems. In large metropolitan environments, the spatial-temporal heterogeneity and dynamic distribution of cases render fixed-area disinfection unable to match outbreak progression, resulting in low resource utilization, delayed coverage, and poor coordination efficiency [1–3]. With the development of smart-city concepts and intelligent epidemic-control paradigms, air-ground-human integrated disinfection systems based on unmanned aerial vehicles (UAVs), ground vehicles, and human workers have gradually become an important direction for improving the precision and responsiveness of epidemic prevention [4–6]. Specifically, existing epidemic disinfection scheduling studies have investigated static or stage-wise optimization strategies based on predefined zones, fixed demand estimation, or single-period resource allocation [7, 8]. Although these approaches achieve certain improvements in coverage efficiency and workload balance, they generally neglect the spatiotemporal evolution of case density

and fail to establish a closed-loop linkage between real-time risk perception and task rescheduling. In recent years, advanced intelligent paradigms such as distributed systems, federated learning, and data-driven decision-making have emerged as promising tools for improving coordination and efficiency in large-scale complex systems [9, 10].

From a control and motion-planning perspective, recent studies have demonstrated the effectiveness of advanced optimization-based and learning-enhanced control strategies in improving the robustness and safety of autonomous systems operating in complex environments [11]. For instance, Li et al. [12] proposed a model predictive optimization framework combined with terminal sliding mode control to ensure accurate motion tracking and obstacle avoidance for mobile robots under system uncertainties. Similarly, hybrid learning-optimization control paradigms have been introduced to coordinate multiple robotic agents in cooperative transportation tasks, highlighting the advantages of combining data-driven adaptability with model-based stability guarantees [13]. These works provide valuable theoretical foundations for high-reliability motion execution in epidemic prevention scenarios. However, their primary focus lies in low-level motion control and cooperative manipulation, without explicit consideration of large-scale task scheduling driven by dynamically evolving risk distributions.

In recent years, significant progress has also been made in multi-agent collaboration and task planning. Methods such as multi-UAV task allocation [14], multi-robot cooperative optimization [15], and integrated air–ground emergency scheduling [16] have provided important references for the automation and intelligence of public health operations. Related path planning studies in UAV-enabled service systems have also explored efficiency-oriented planning under resource constraints [17, 18]. At the planning level, real-time whole-body motion planning frameworks, such as the RAMPAGE system, have shown strong capabilities in enabling mobile manipulators to operate agilely in unknown and cluttered environments through unified perception-planning-control pipelines [19]. Meanwhile, advances in lightweight and robust simultaneous localization and mapping (SLAM), exemplified by Robust Lightweight Dynamic SLAM, enhance autonomous perception reliability in dynamic environments via joint semantic and motion information exploitation [20]. These studies significantly enhance the autonomy and environmental adaptability of robotic systems, which are essential for large-scale urban disinfection tasks.

Nevertheless, most existing approaches still rely on predefined tasks or uniform area partitioning strategies and lack dynamic priority modeling mechanisms explicitly driven by case density and spatiotemporal risk evolution. Moreover, the majority of current systems adopt open-loop or weakly coupled planning-execution architectures. The absence of closed-loop feedback mechanisms and rolling revision strategies during task execution often leads to delayed responses and imbalanced scheduling when facing unexpected disturbances such as equipment failures, lockdown adjustments, or disinfectant shortages [21, 22]. Therefore, how to construct a case-density-driven, closed-loop intelligent scheduling framework that tightly integrates perception, risk assessment, task allocation, and execution adjustment remains a critical challenge for air-ground-human collaborative epidemic prevention systems.

To address these challenges, this study proposes a case-density-driven, closed-loop intelligent disinfection strategy integrating air, ground, and human resources, and constructs an end-to-end framework linking case perception with task scheduling. By fusing case data and population mobility information, the framework constructs a real-time updated spatiotemporal risk field, enabling adaptive identification and prioritization of high-risk regions. Based on a multi-agent collaboration model, a multi-objective optimization framework is established, focusing on coverage efficiency, suppression timeliness, path conflict minimization, and resource cost constraints. Furthermore, a closed-loop control process is formed by incorporating execution feedback and rolling re-planning, achieving dynamic adjustment across task allocation, execution monitoring, and strategy refinement. The main goals of this study are as follows:

(1) To achieve risk self-perception and priority-driven scheduling based on dynamically evolving case distributions.

(2) To establish a multi-objective optimization scheduling model for air-ground-human collaborative disinfection tasks.

(3) To enhance system robustness and operational stability under complex disturbances through a closed-loop rolling mechanism.

The main innovations and contributions of this paper are as follows:

(1) A case-density-driven method for public health disinfection task prioritization and risk modeling is proposed, enabling dynamic construction and continuous updating of the risk field.

(2) A multi-objective optimization model for air-ground-human collaborative task scheduling is designed, balancing efficiency, coverage, and resource constraints.

(3) A closed-loop scheduling strategy combining execution feedback with rolling re-planning is developed, achieving adaptive control and robust optimization of the system.

(4) A series of multi-scenario experiments and comparative analyses validate the advantages of the proposed method in operational efficiency, coverage rate, and disturbance resilience.

The findings of this study provide theoretical support and technical references for achieving intelligent, precise, and highly resilient public health disinfection operations.

2 System Modeling and Problem Formulation

2.1 Risk Field Modeling

In the case-density-driven air-ground-human collaborative disinfection system, the accurate construction of a risk field serves as the foundation for task planning and resource allocation. The risk field characterizes the infection transmission risk and its temporal evolution across different spatial regions, enabling dynamic identification of high-risk areas and quantification of their priority levels through joint analysis of case distribution and population mobility.

2.1.1 Case data processing and spatiotemporal distribution modeling

Assume that during the time period $T = [t_0, t_N]$, the city is divided into $M \times N$ grid cells, and the center coordinate of each-grid cell is denoted as (x_i, y_i). The reported case data can be represented as $D = \{(x_k, y_k, t_k, w_k) \mid k = 1, 2, \ldots, K\}$. (x_k, y_k) is the spatial coordinate of case k, t_k is the reporting time of the case, and w_k is the weight of case mmm (for example, confirmed cases are 1, suspected cases are 0.5).

To reflect the diffusion influence of cases in space, Gaussian Kernel Density Estimation is used to construct the case density field $\rho(x, y, t) = \sum_{k=1}^{K} w_k \cdot \left(-\frac{(x-x_k)^2+(y-y_k)^2}{2\sigma_s^2} - \frac{(t-t_k)^2}{2\sigma_t^2}\right)$. $\rho(x, y, t)$ is the case density at time t and position (x, y); σ_s is the spatial diffusion scale parameter, used to control the influence range of cases; σ_t is the time decay parameter, reflecting the attenuation speed of case influence over time.

2.1.2 Integration of population mobility information

Case distribution alone is insufficient to accurately describe potential transmission risks; therefore, population mobility information $F(x, y, t)$ is incorporated to construct a mobilityweighted risk field. Population mobility intensity can be derived from cellular signaling data or traffic monitoring data, formally defined as $F(x, y, t) = \frac{1}{A}\sum_{n-1}^{N_p} v_n(t) \cdot \delta\left(x - x_n(t), y - y_n(t)\right)$. A is the area of grid cell; N_p is the number of individuals n within the time window; $v_n(t)$ denotes the movement speed or travel frequency of individual n; $(x_n(t), y_n(t))$ is the position of individual n and $\delta\left(x - x_n(t), y - y_n(t)\right)$ is the Dirac delta function used to indicate the spatial location of each individual.

By combining the case density field and population mobilityfield, a composite risk field is constructed as $R(x, y, t) = \alpha \cdot \rho(x, y, t) + \beta \cdot \frac{F(x, y, t)}{\max(F(x, y, t))}$. α and β are weighting coefficients reflecting the contributions of case distribution and population mobility, respectively; and $R(x, y, t)$ represents the overall risk intensity at location (x, y) and time t.

2.1.3 Priority scoring model for high-risk regions

For subsequent task scheduling and resource assignment, the risk field must be transformed into priority scores for discrete spatial regions. Let the region partition set be denoted as $\Omega = \{A_1, A_2, \ldots, A_L\}$. The priority score S_i of region A_i is defined as:

$$S_i = \int_{A_i} \left[\lambda_1 R(x, y, t) + \lambda_2 \nabla^2 R(x, y, t)\right] dxdy \tag{1}$$

where, S_i is the priority score of region A_i; λ_1 and λ_2 are weighting coefficients controlling the contributions of risk intensity and spatial gradient, respectively; and $\nabla^2 R(x, y, t)$ represents the Laplacian operator of the risk-field, capturing the rate of risk variation (regions with sharp risk gradients indicate potential transmission boundaries).

Finally, through normalization, the standardized priority score $\tilde{S}_i \in [0, 1]$ is obtained and used in subsequent disinfection task planning and scheduling algorithms $\tilde{S}_i = \frac{S_i - \min(S)}{\max(S) - \min(S)}$.

2.2 Multi-Agent Collaborative Model

To achieve efficient collaboration among UAVs, ground vehicles, and operators in public health disinfection tasks, it is necessary to establish a model of the operational characteristics of multiple agents and their mutual constraints. This model is used to describe the kinematic characteristics, operational capabilities, and resource limitations of various agents during execution, providing the basis for subsequent scheduling optimization.

2.2.1 Unmanned aerial vehicles platform model

Let the set of UAVs be $U = \{u_1, u_2, \ldots, u_{N_u}\}$. The state of each UAV u_i can be expressed as $P_i^u(t) = \{x_i^u(t), y_i^u(t), v_i^u(t), e_i^u(t), l_i^u(t)\}$. $(x_i^u(t), y_i^u(t))$ is the spatial position of UAV u_i at time t; $v_i^u(t)$ is the flight speed; $e_i^u(t)$ is the remaining battery level; and $l_i^u(t)$ is the current amount of carried disinfectant.

The motion constraint model of the UAV is defined as $\dot{x}_i^u(t)^2 + \dot{y}_i^u(t)^2 = \left(v_i^u(t)\right)^2, 0 < v_i^u(t) < v_{\max}^u$. $v_{\max}^u$ is the maximum flight speed. The endurance time of the UAV is constrained by battery level and defined as $e_i^u(t + \Delta t) = e_i^u(t) - \eta_e^u \cdot v_i^u(t) \cdot \Delta t$. η_e^u is the energy consumption coefficient, representing the power consumption rate per unit flight distance. The dynamic change of disinfectant amount can be expressed as $l_i^u(t + \Delta t) = l_i^u(t) - \eta_l^u \cdot A_i^u(t)$.

$A_i^u(t)$ is the spray coverage area of UAV u_i at time t; η_l^u is the disinfectant consumption per unit area. When $e_i^u(t) \le e_{\min}^u$ or $l_i^u(t) \le l_{\min}^u$, the UAV must return to the supply station for charging or refilling.

2.2.2 Ground vehicle and personnel model

Let the set of ground vehicles be $V = \{v_1, v_2, \ldots, v_{N_v}\}$, and the set of personnel be $H = \{h_1, h_2, \ldots, h_{N_h}\}$. The motion state of a ground vehicle is $P_j^v(t) = \left\{x_j^v(t), y_j^v(t), s_j^v(t), c_j^v(t)\right\}$. $s_j^v(t)$ is the driving speed; $c_j^v(t)$ is the current disinfectant load or operational capability; $\left(x_j^v(t), y_j^v(t)\right)$ is the vehicle position; and the driving constraint is $0 < s_j^v(t) < s_{\max}^v$.

The personnel model mainly focuses on coverage and time constraints, that is, $C_k^h(t) = \left\{x_k^h(t), y_k^h(t), r_k^h, T_k^{\text{work}}\right\}$. r_k^h is the effective operation radius of personnel; T_k^{work} is the continuous operation time; and $\left(x_k^h(t), y_k^h(t)\right)$ is the current position of personnel. To ensure the efficiency of ground collaborative operations, the coverage interval of personnel must satisfy $\bigcup_{k=1}^{N_h} \mathrm{A}_k^h(t) \subseteq \mathrm{A}_{\text{task}}$, that is, the coverage regions of all personnel must include the task area.

2.2.3 Collaborative constraint conditions

During task execution, aerial, ground, and human agents must satisfy task dependency constraints, spatial conflict avoidance constraints, and resource limitation constraints.

For task dependency constraints, disinfection tasks are executed according to priority based on risk. If region A_i has precedence constraints(e.g., ground personnel must first implement control before UAV spraying), then $T_{\text{start}}^{u_i} \ge T_{\text{end}}^{h_j}, \forall (A_i, A_j) \in D_{\text{dep}}$. D_{dep} denotes the task dependency set.

For spatial conflict avoidance constraints, UAVs and vehicles performing tasks simultaneously must maintain a safe distance $\left\|P_i^u(t) - P_j^n(t)\right\| \ge d_{\min}^{uv}, \forall t$. $d_{\min}^{uv}$ is the minimum air-ground safety distance. Agents of the same type must also satisfy the $d_{\min}^{uu}$ and $d_{\min}^{uv}$ constraints.

For resource limitation constraints, the total system resources satisfy $\sum_i l_i^u(t) + \sum_j c_j^v(t) + \sum_k r_k^h \le L_{\text{total}}$. L_{total} is the total amount of disinfectant resource in the system. Considering task completion rate and energy consumption constraints, $\frac{\sum_{A_i \in \Omega_c} S_i}{\sum_{A_i \in \Omega} S_i} \ge \theta$, $E_{\text{total}} \le E_{\max}$. Ω_c is the set of completed regions; S_i is the priority score of region, θ is the minimum task completion ratio. E_{total} and $E_{\max}$ represent the total system energy consumption and its upper limit, respectively.

Through the above model definitions and constraint descriptions, the multi-agent collaboration problem can be abstracted into a multi-objective scheduling optimization problem, whose goal is to maximize task coverage and operation efficiency while minimizing energy consumption and time delay under resource and safety constraints, providing a modeling foundation for subsequent optimization algorithm design and dynamic scheduling.

2.3 Problem Definition

Based on the integrated risk field modeling and multi-agent collaboration model, this study defines the air-ground-human collaborative disinfection task planning problem as a multi-objective optimization problem. Its core goal is to achieve collaborative control that maximizes task coverage, minimizes suppression time, optimizes operational efficiency, reduces costs, and enhances system robustness under limited resource constraints.

2.3.1 Optimization objectives

Let the system scheduling scheme be $\Pi = \{\pi_1, \pi_2, \ldots, \pi_T\}$. Each sub-strategy π_t corresponds to the multi-agent task allocation and path decision at time t. The multi-objective optimization function can be defined as: $\min_\Pi F(\Pi) = \{f_1(\Pi), f_2(\Pi), f_3(\Pi), f_4(\Pi), f_5(\Pi)\}$. Operational Efficiency (OE, $f_1(\Pi)$) measures the number of effective tasks completed per unit time $f_1(\Pi) = -\frac{1}{T}\sum_{t=1}^{T} \frac{|\Omega_c(t)|}{|\Omega|} |\Omega_c(t)|$ is the number of task areas completed at time t. The negative sign indicates maximization of this metric.

Suppression Time (ST, $f_2(\Pi)$) quantifies the delay from risk identification to task completion $f_2(\Pi) = \frac{1}{|\Omega|}\sum_{A_i \in \Omega} (T_{\text{end},\, i} - T_{\text{risk},\, i})$. $T_{\text{risk},\, i}$ is the time when area A_i is identified as high-risk, and $T_{end,\, i}$ is the completion time of the disinfection task in that area.

Coverage Ratio (CR, $f_3(\Pi)$) reflects the proportion of the task area effectively covered, $f_3(\Pi) = -\frac{\sum_{A_i \in \Omega_c} S_i}{\sum_{A_i \in \Omega} S_i}$, S_i is the risk-priority score of area A_i. The negative sign indicates maximization of coverage.

Resource Cost (RC, $f_4(\Pi)$) reflects total energy and disinfectant consumption $f_4(\Pi) = \sum_{i=1}^{N_u} (\kappa_e^u E_i^u + \kappa_l^u L_i^u) + \sum_{j=1}^{N_v} \left(\kappa_e^v E_j^v + \kappa_l^v L_j^v\right)$. E_i^u and E_j^ν are energy consumptions of UAVs and vehicles, L_i^u and L_j^v are disinfectant consumptions, and the coefficients represent energy and disinfectant cost factors.

System Robustness (SR, $f_5(\Pi)$) measures the system's adaptability to unexpected disturbances (e.g., equipment failure, path blockage, weather changes) $f_5(\Pi) = \frac{1}{T}\sum_{t=1}^{T} \left(1 - \frac{|\Delta\Pi_t|}{|\Pi_t|}\right)$. $\Delta\Pi_t$ is the number of tasks requiring replanning due to disturbances. A larger value indicates stronger robustness.

A composite optimization function can then be constructed using a weighted normalization form $\min_\Pi F_{\text{total}} = \sum_{m=1}^{5} w_m f_m(\Pi)$. $w_m \in [0, 1]$ is the weight coefficient representing the relative importance of each objective.

2.3.2 Constraints

The multi-objective optimization problem must satisfy the following constraints.

Resource Constraint: $\sum_i l_i^u(t) + \sum_j c_j^v(t) \leq L_{\text{total}}, \forall t$, indicating limited total disinfectant availability.

Time Constraint: $T_{\text{end},i} - T_{\text{start},i} \leq T_{\text{total}}, \forall A_i \in \Omega$, i.e., the duration of a single task cannot exceed the maximum allowed time.

Task Feasibility Constraint: A task is executed only if resource, distance, and safety conditions are satisfied; otherwise, it is not executed. Tasks are schedulable only when energy, payload, and conflict conditions are met.

Collaboration and Safety Constraint: $\left\|P_i^u(t) - P_j^v(t)\right\| \geq d_{\min}^{uv}$, $\|P_i^u(t) - P_k^u(t)\| \geq d_{\min}^{uu}$, ensuring safe distances among aerial, ground, and human agents during execution.

2.3.3 Closed-loop control requirement

Since the risk field and task status dynamically change, the system requires closed-loop control.

Risk Feedback Perception: Real-time monitoring of case distribution, human mobility, and execution status to update the risk field Π_t.

Dynamic Task Adjustment: Real-time modification of task priorities, allocation schemes, and path planning based on feedback results.

Rolling Optimization Execution: Using Model Predictive Control or reinforcement learning strategies to optimize SSS in a rolling time horizon.

Disturbance Adaptation: Upon detecting sudden events (e.g., UAV failure, area lockdown), the system automatically triggers local replanning: $\Pi(t + \Delta t) = \Pi(t) + \eta \cdot \Delta F(\Pi(t))$. η is the adjustment step size, and $\Delta F(\Pi(t))$ is the change in the objective function gradient guiding the scheduling strategy to convergence.

Through this closed-loop mechanism, the system can maintain continuous, adaptive execution of optimal disinfection strategies under dynamic risk conditions, achieving collaborative closed-loop control among air, ground, and human agents.

2.4 Closed-Loop Intelligent Strategy Design

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2.4.1 Intelligent task allocation model

To achieve dynamic multi-agent (air-ground-human) collaborative disinfection scheduling, this study first constructs a dynamic input layer of regional risk distribution based on realtime case reports and human mobility data. Let the study area be divided into Ω grid cells $\{\omega_i \mid i = 1, 2, \ldots, N\}$; the risk value of each grid cell at time t is defined as $R_i(t) = \alpha \cdot C_i(t) + \beta \cdot F_i(t)$. $R_i(t)$ is the comprehensive risk value of the i th grid cell; $C_i(t)$ is the case density (number of cases/population) per unit time; $F_i(t)$ represents population inflow intensity, calculated from mobile communication or travel trajectory data; α and β are weight coefficients reflecting the relative influence of case data and population movement on risk assessment.

To mitigate short-term fluctuation effects, a time-weighted smoothing mechanism is introduced $\tilde{R}_i(t) = (1 - \lambda) \cdot R_i(t) + \lambda \cdot \tilde{R}_i(t-1)$. $\lambda \in [0, 1]$ is a time decay factor balancing the weights of current data and historical trends. A larger λ makes the system more sensitive to historical changes, suitable for long-term trend monitoring, while a smaller λ emphasizes short-term response.

Based on real-time risk calculation, a spatially continuous risk field function $R(x, y, t)$ is constructed and dynamically updated in a rolling prediction manner, providing a basis for subsequent task priority allocation. Spatial interpolation of the risk field uses a Gaussian diffusion model $R(x, y, t) = \sum_{i=1}^{N} \tilde{R}_i(t) \cdot \exp\left(-\frac{(x-x_i)^2+(y-y_i)^2}{2\sigma^2}\right)$. (x_i, y_i) is the center coordinate of grid cell ω_i, and σ is the spatial diffusion coefficient controlling the spatial range of risk influence.

Considering the continuity of epidemic spread and population mobility, the risk field is updated in a sliding time window $[t, t + \Delta t]$: $R(x, y, t + \Delta t) = (1 - \mu)R(x, y, t) + \mu \cdot R_{\text{new}}(x, y, t + \Delta t)$. $R_{\text{new}}(x, y, t + \Delta t)$ is the risk estimate calculated from the latest case and mobility data, and $\mu \in [0, 1]$ is the rolling update coefficient controlling the system's response speed and stability.

When the epidemic situation changes rapidly, μ can be increased to enhance model sensitivity; when the risk field is relatively stable, μ can be decreased to maintain continuity and robustness. Through this rolling update mechanism, the system can reflect the real-time evolution of the spatial risk pattern, enabling the task allocation model to adaptively respond to sudden risk changes.

2.4.2 Multi-objective task scheduling optimization

(1) Choice of optimization method

Due to the multi-objective conflicts (efficiency, coverage, energy consumption, robustness, etc.) and combinatorial complexity (task allocation, path planning, time synchronization) inherent in air-ground-human multi-agent collaborative task scheduling, traditional single-method algorithms struggle to obtain globally feasible solutions within reasonable time. This study adopts a hybrid heuristic multi-objective optimization approach, combining an improved genetic algorithm (Improved NSGA-II) with a reinforcement learning strategy.

Specifically, the genetic algorithm searches globally for candidate solutions of task allocation and path combinations, quickly converging toward the Pareto front; the reinforcement learning module, in each iteration, uses environmental feedback (e.g., coverage improvement, task conflict reduction) to locally adapt each individual, enhancing the dynamic optimality and stability of global solutions. This hybrid mechanism achieves a superior exploration–exploitation balance, maintaining genetic search diversity while enabling real-time self-optimization through policy updates.

(2) Multi-objective coupled modeling

Based on the rolling update of the risk field, the scheduling problem is formalized as a multi-objective optimization model $\min_X F(X) = \{f_1(X), f_2(X), f_3(X), f_4(X), f_5(X)\}$. The decision variable $X = \{x_{ij}, p_k, t_l\}$ represents the task allocation matrix, path decisions, and time scheduling vectors. The objective functions are defined as follows.

Operational Efficiency Maximization: expressed as the area covered per unit time: $f_1 = -\frac{1}{T}\sum_{i=1}^{M}\sum_{j=1}^{N} a_{ij}x_{ij}$. a_{ij} is the coverage area of agent i executing task j, and $x_{ij} \in [0, 1]$ indicates whether the task is allocated. The negative sign denotes a minimization formulation.

Suppression Time Minimization: representing the average delay from risk identification to task completion $f_2 = \frac{1}{N}\sum_{j=1}^{N}(t_{\text{end},\,j} - t_{\text{sart},\,j})$.

Energy Cost Minimization: including UAV flight, ground vehicles, and personnel operational energy consumption $f_3 = \sum_{i=1}^{M}(E_i^u + E_i^v + E_i^p)$.

Coverage Maximization: measuring the proportion of highrisk areas successfully covered $f_4 = -\frac{\sum_{\omega_i \in \Omega_H} c_i^{\text{done}}}{\sum_{\omega_i \in \Omega_H} 1}$. Ω_H is the set of high-risk grid cells, and c_i^{done} is the set of cells already covered.

Robustness Maximization: reflecting the ability to maintain tasks under unexpected disturbances $f_4 = -\operatorname{Var}\left(\frac{\Delta R}{\Delta t}\right)$. $\frac{\Delta R}{\Delta t}$ represents fluctuations in risk reduction rate; a smaller variance indicates smoother system response.

(3) Constraint handling and feasible solution generation

To ensure the executability of scheduling results, the model must satisfy the following constraints.

Resource Constraints: the task number and payload of each agent type are limited $\sum_{j=1}^{N} x_{ij} \cdot \theta_j \le W_i^{\max}, \forall i$. θ_j is the chemical demand for task j, and $W_i^{\max}$ is the maximum payload of agent i.

Time Constraints: all tasks must be completed within the allowed time window $\left[t_j^{\min}, t_j^{\max}\right]$, $t_{\text{start},\,j} \ge t_j^{\min}, t_{\text{end},\,j} \le t_j^{\max}$.

Spatial Conflict Constraints: no overlapping operations by multiple agents in the same area at the same time $\forall\,(\omega_m, t) \in \Omega,\ \sum_i \delta_{i,\,m}(t) \le 1$.

Task Dependency Constraints: ground personnel tasks can only be executed after UAV spraying of upper-level areas is completed $x_{g,\,j} \le x_{a,\,j}, \forall j$. Feasible solutions are generated using a two-level "constraint repair + penalty function" mechanism: first, clearly conflicting solutions are eliminated during genetic crossover and mutation; second, minor violations (e.g., time window exceedance, insufficient payload) are penalized via a penalty function $\Phi(X) = \sum_{c=1}^{C} \eta_c \cdot \text{viol}_c(X)$. η_c is the constraint weight and viol $_c(X)$ quantifies the degree of violation for constraint class ccc. The final optimization objective is converted into a weighted composite form $\min F^*(X) = F(X) + \Phi(X)$.

This constraint-handling strategy ensures diversity in the global search space while maintaining feasibility during convergence, laying the foundation for subsequent closed-loop scheduling and dynamic feedback optimization.

2.4.3 Execution closed-loop and adaptive replanning

(1) Task execution state monitoring

To realize collaborative closed-loop operation of air-ground-human multi-agents, the system establishes a multisource task execution state monitoring module that collects realtime UAV flight status, ground vehicle progress, and personnel trajectory information. The execution state vector of each task T_i is defined as $S_i(t) = [p_i(t), v_i(t), e_i(t), r_i(t)]$. $p_i(t)$ represents the current position coordinates; $v_i(t)$ denotes velocity and operational posture; $e_i(t)$ is energy consumption and equipment status (e.g., UAV battery level); $r_i(t)$ represents task completion ratio (coverage rate).

State data are reported via wireless communication links, and the scheduling center periodically integrates all agent states to generate the system execution state matrix $S(t)$, providing dynamic input for rolling optimization. If

task execution anomalies or deviations are detected (e.g., energy consumption exceeds threshold, coverage progress falls below expectation), the system triggers local replanning.

(2) Disturbance handling mechanism

In complex public health scenarios, task execution may be affected by various unexpected disturbances, mainly including:

Sudden lockdown areas: some high-risk areas are temporarily restricted due to outbreak spread, preventing originally planned tasks from being executed;

Insufficient chemical supply: UAVs or vehicles may run out of chemicals and need to return to resupply points, causing task delays;

UAV or vehicle failures: equipment anomalies (loss of connection, motor faults, etc.) require rapid replacement of executing agents.

To address these, a disturbance-aware state transition mechanism is designed. Let the system's current optimal schedule be $X^*(t)$; upon detecting a disturbance event $\delta(t)$, the system automatically selects a recovery strategy $X(t+\Delta t) = \begin{cases} X^*(t) \\ F_{\text{local}}\left(X^*(t), \delta(t)\right) \\ F_{\text{glocal}}\left(X^*(t), \delta(t)\right) \end{cases}$. F_{local} is a local replanning function that adjusts only the affected task subset, and F_{glocal} is a global re-optimization function used for major disturbances (e.g., UAV loss, large-scale lockdown).

(3) Rolling replanning algorithm frameworkm

Rolling Replanning is the core of closed-loop intelligent scheduling. The system checks task states and risk field changes at fixed periods Δt; if deviations exceed a threshold $\mathcal{E}$, the reoptimization process is triggered. The basic algorithm flow is as follows:

Input: current scheduling solution $X^*(t)$, state matrix $S(t)$, risk field $R(x, y, t)$;

Disturbance detection: if deviation $|S(t) - S_{\text{pred}}(t)| > \varepsilon$, mark abnormal task set T_{abn};

Local update: re-evaluate priorities for tasks in T_{abn} and generate candidate set T';

Local optimization: execute hybrid heuristic algorithm H to obtain corrected solution $X'(t)$;

Strategy fusion: merge old solution and new solution via weighted averaging: $X(t+\Delta t) = (1-\zeta)X(t)+\zeta X'(t)$. ζ is the strategy update coefficient;

Output: updated executable scheduling plan $X(t + \Delta t)$.

This mechanism ensures continuous scheduling and global stability under task disturbances and evolving risk conditions.

(4) Task demand prediction and dynamic scheduling model

To enhance system foresight and resource utilization efficiency, a task demand prediction model based on case trends is introduced. Let the case density vectors over the past k time windows be $C_{t-k,\,t}$; future risk increment is predicted via a time-series regression model $\tilde{C}(t + \Delta t) = \gamma_0 + \sum_{i=1}^{k} \gamma_i \cdot C(t - i\Delta t) + \xi(t)$. γ_i is the historical weight coefficient reflecting the lagged impact of case changes; $\xi(t)$ is a disturbance term modeling sudden factors. Prediction results are used to update the risk field $R(x, y, t+\Delta t)$, preemptively triggering task resource preallocation and path pre-generation, enabling prediction-driven dynamic scheduling.

This mechanism equips the system with autonomous early warning and proactive response capabilities in rapidly evolving epidemic scenarios, significantly enhancing overall operational stability and timeliness.

3 Experiments & Performance Evaluation

3.1 Experimental Setup

To verify the effectiveness of the proposed case-ensity-riven air-ground-human closed-loop intelligent disinfection strategy, this study establishes a comprehensive experimental platform to simulate urban public health scenarios under multiagent collaboration, dynamic risk evolution, and task disturbance conditions. The experimental environment is based on the spatial structure of a typical urban area, with a city space model constructed using grids of approximately 1000 m $\times$ 1000 m. Each grid contains information on population density, road network, and building distribution, providing the foundation for risk field modeling and coverage calculation.

The multi-agent modules include UAVs, ground vehicles, and personnel, each with engineering-realistic attributes such as operational speed, chemical capacity, flight/work duration, and operational radius. They also support real-time feedback on position, speed, and energy consumption. The system employs a rolling optimization platform to implement dynamic risk field updates, task allocation, multi-objective optimization, closed-loop scheduling, and disturbance handling, automatically triggering local or global replanning to simulate real-world uncertainties such as equipment failure, lockdown changes, and chemical depletion.

Regarding experimental data, case information is based on publicly reported epidemic time series and spatial clustering characteristics of a real city. Spatial density distributions of cases are simulated using a Poisson intensity function, and case trend evolution is modeled via an autoregressive approach. Exogenous clustering disturbances are added to simulate sudden outbreak events, making the risk field exhibit more realistic nonlinear dynamics. Population

mobility data are derived from publicly available origin-destination travel statistics, incorporating commuting flows, commercial activity flows, and random movement patterns, which are mapped onto the spatial grid to update population exposure in each region in real time. Case density and mobility intensity are linearly weighted to form the dynamic risk field input used in experiments.

To comprehensively evaluate system performance, three typical test scenarios are designed.

Scenario A: dense central urban area with complex road networks, high population mobility, and clustered case distribution. This scenario tests scheduling efficiency and coverage blind spot suppression under high load and complexity.

Scenario B: mixed old residential and market areas, with irregular structure and multi-cluster scattered cases. This scenario examines cross-platform coordination of air-ground-human agents and task balancing under complex boundary conditions.

Scenario C: large open public activity areas (e.g., squares, parks), with highly fluctuating population flow and randomly occurring case peaks, suitable for evaluating UAV rapid coverage, response speed, and robustness under disturbance.

By combining these scenarios, a comprehensive performance evaluation system is established, ensuring that the proposed intelligent disinfection framework can adapt to diverse urban structures, multiple case propagation patterns, and multi-agent collaborative environments.

3.2 Comparison Methods

To evaluate the performance of the proposed closed-loop air-ground-human collaborative disinfection strategy, four representative baseline methods are selected for comparison.

Static scheduling: traditional baseline where disinfection paths and task allocation are generated before task commencement based on fixed rules or historical experience, without updates according to case density, population flow, or equipment status. While simple, this method may exhibit scheduling delays, low coverage efficiency, and severe resource waste under rapidly changing epidemics, serving as a reference for dynamic scheduling advantages.

Single-platform planning: structured comparison using only UAVs or only ground teams for task planning. This demonstrates local optimality of a single agent but lacks cross-platform coordination, failing to balance rapid aerial coverage with fine-grained ground operations, thus performing poorly in multi-scale, complex terrain, or highly dynamic risk scenarios.

Traditional multi-objective scheduling: baseline optimization using classical multi-objective genetic algorithms (NSGA-II), decomposition-based evolutionary methods, or multi-objective constraint programming tools. These methods can optimize efficiency, coverage, and cost, but typically rely on static cost functions, lacking case-density-driven mechanisms and task execution feedback loops, limiting adaptation to dynamic risk fields and changing resource states.

Proposed case-density-driven closed-loop method: the core approach of this study. It constructs a dynamic risk field from real-time case data, population mobility, and platform status, and uses multi-agent collaboration modeling for coordinated scheduling of UAVs, ground vehicles, and personnel. Combined with closed-loop control and rolling replanning, the system can respond to unexpected events (e.g., lockdown adjustments, equipment failures, chemical depletion) in real time. The end-to-end perception-evaluation-scheduling-execution framework provides superior real-time performance, robustness, and public health protection capabilities.

These four methods span from static to dynamic, single-agent to multi-agent, and open-loop to closed-loop feedback, forming a hierarchical evaluation system to comprehensively demonstrate the advantages of the proposed method in complex public health disinfection tasks.

3.3 Performance Metrics

To quantitatively assess the performance of the proposed case-density-driven air-ground-human closed-loop collaborative disinfection method, five core metrics are designed.

Coverage rate (CR): measures the proportion of high-risk areas and critical population pathways covered during the task cycle, reflecting the match between operational strategy and the risk field. Higher CR indicates more effective intervention along potential transmission paths.

Time-to-control (TTC): measures the time required to suppress overall risk field intensity to safe levels from task initiation. TTC is directly related to case density decay, disinfection response speed, and scheduling strategy alignment. Lower TTC indicates faster containment of potential transmission chains.

Work efficiency (WE): reflects effective disinfection area or task quantity completed per unit time, encompassing path optimization, platform operational efficiency, and collaborative effects. This metric compares execution efficiency across methods, particularly highlighting improvements from air-ground-human coordination.

Resource utilization (RU): evaluates how fully UAVs, ground vehicles, and personnel are utilized during task execution. It accounts for task allocation balance and whether cross-platform collaboration avoids idle or overloaded resources. Higher RU indicates efficient multi-agent system usage.

Task balance (TB): measures the balance of task allocation among UAVs, vehicles, and personnel in terms of workload, work duration, and coverage pressure. Higher TB indicates more equitable distribution, reducing risks of equipment failure, fatigue, or task delays, and reflecting system-level scheduling optimization.

Robustness index (RI): assesses the system's ability to maintain task performance under uncertainty and disturbances (e.g., equipment failure, chemical shortage, temporary lockdown changes). Disturbance experiments quantify performance degradation and recovery speed. Higher RI indicates stronger adaptive adjustment and risk resistance.

Together, these six metrics provide a multi-dimensional evaluation framework, comprehensively demonstrating the reliability, efficiency, and practical value of scheduling strategies in public health disinfection tasks.

3.4 Experimental Results Analysis

3.4.1 Comparison of metrics under different strategies

To comprehensively evaluate scheduling strategies for public health disinfection tasks, this study compares four methods: static scheduling, single-platform planning, traditional multi-objective scheduling, and the proposed closed-loop air-ground-human collaborative strategy.

Across the three typical scenarios—scenario A, scenario B, and scenario C—the mean values and standard deviations of six performance metrics for each method are presented in Table 1, 2, and 3.

Table 1. Performance comparison under scenario A (dense urban area)

Method	CR	TTC(min)	WE(ha/h)	RU(%)	TB	RI
Static scheduling	0.63 ± 0.05	242.35 ± 28.47	5.12 ± 0.78	55.32 ± 5.88	0.44 ± 0.06	0.36 ± 0.05
Single-platform planning	0.69 ± 0.04	185.72 ± 23.16	6.48 ± 0.87	59.83 ± 4.79	0.41 ± 0.06	0.32 ± 0.05
Traditional multi-objective scheduling	0.77 ± 0.04	124.56 ± 14.32	7.92 ± 0.71	71.24 ± 3.92	0.63 ± 0.05	0.59 ± 0.06
Proposed method	0.87 ± 0.03	63.48 ± 10.57	10.54 ± 0.92	83.67 ± 2.85	0.81 ± 0.04	0.88 ± 0.03

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

Table 2. Performance comparison under scenario B (mixed old community & market area)

Method	CR	TTC(min)	WE(ha/h)	RU(%)	TB	RI
Static scheduling	0.61 ± 0.05	238.47 ± 27.88	5.03 ± 0.79	54.87 ± 5.91	0.43 ± 0.07	0.35 ± 0.06
Single-platform planning	0.70 ± 0.04	178.92 ± 23.47	6.72 ± 0.88	60.12 ± 4.85	0.42 ± 0.06	0.33 ± 0.05
Traditional multi-objective scheduling	0.78 ± 0.04	119.48 ± 14.72	8.03 ± 0.72	72.13 ± 3.91	0.66 ± 0.05	0.61 ± 0.06
Proposed method	0.88 ± 0.03	61.32 ± 10.23	10.78 ± 0.95	84.25 ± 2.91	0.83 ± 0.04	0.87 ± 0.03

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

Table 3. Performance comparison under scenario C (large public activity area)

Method	CR	TTC(min)	WE(ha/h)	RU(%)	TB	RI
Static scheduling	0.62 ± 0.05	236.89 ± 29.12	5.08 ± 0.81	55.12 ± 6.02	0.44 ± 0.06	0.36 ± 0.06
Single-platform planning	0.70 ± 0.04	182.34 ± 24.56	6.55 ± 0.90	59.89 ± 4.92	0.40 ± 0.06	0.31 ± 0.05
Traditional multi-objective scheduling	0.78 ± 0.04	121.73 ± 14.89	8.08 ± 0.70	71.88 ± 3.95	0.65 ± 0.05	0.60 ± 0.06
Proposed method	0.88 ± 0.03	62.47 ± 10.45	10.65 ± 0.98	83.92 ± 2.87	0.82 ± 0.04	0.87 ± 0.03

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

To further analyze the adaptability and performance variations of different scheduling strategies across scenarios, this study examined the trends of six performance metrics across three typical scenarios, as shown in Figure 1.

In terms of CR, all methods exhibited some fluctuation across the three scenarios. Static scheduling and single-platform planning showed minor increases or decreases, ranging from 0.61 to 0.70, indicating that their coverage capability is strongly affected by environmental complexity and case distribution. Traditional multi-objective scheduling maintained a stable CR between 0.77 and 0.78, demonstrating certain adaptability to scenario changes through global optimization. The proposed closed-loop air-ground-human collaborative strategy consistently

achieved high coverage rates (0.87–0.88) across all scenarios with minimal fluctuation, indicating effective adaptation to varying case distribution patterns and rapid coverage of high-risk areas.

Regarding TTC, static scheduling and single-platform planning showed considerable variation across scenarios, ranging from 178 to 242 minutes. Traditional multi-objective scheduling was relatively stable (119–125 minutes). In contrast, the proposed method maintained TTC around 61–63 minutes in all scenarios, demonstrating that closed-loop dynamic scheduling, combined with real-time risk field rolling updates and task feedback, can quickly respond to public health disinfection demands under varying environmental conditions.

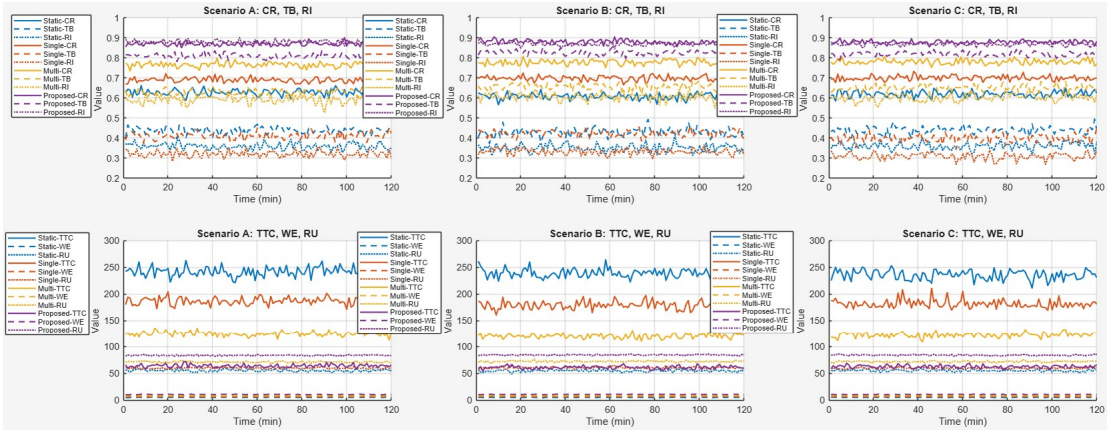


Figure 1. Comparison of six performance metrics for different scheduling methods across three scenarios

For WE and RU, static and single-platform methods were heavily influenced by scenario complexity, with WE ranging from 5.03 to 6.72 ha/h and RU from 54.87% to 60.12%, reflecting inefficient resource utilization under single-platform or static routing. Traditional multi-objective scheduling achieved WE of 7.92–8.08 ha/h and RU of 71.24–72.13% across all scenarios, indicating that global optimization improves efficiency and resource utilization. The proposed strategy reached WE of 10.54–10.78 ha/h and RU of 83.67–84.25%, with minimal variation across scenarios, demonstrating that multi-agent collaboration and dynamic task allocation can stably and efficiently complete disinfection tasks.

Regarding TB, static and single-platform methods showed little difference across scenarios (0.40–0.44). Traditional multi-objective scheduling was slightly higher (0.63–0.66), while the proposed method remained stable at 0.81–0.83, indicating that the closed-loop strategy balances task loads across platforms, reduces the risk of overloading any single platform, and maintains task equilibrium in both dense urban areas and large open spaces.

For System RI, static and single-platform methods were more affected by disturbances (0.31–0.36), traditional multi-objective scheduling achieved RI of 0.59–0.61, and the proposed method remained stable at 0.87–0.88. This demonstrates that the closed-loop strategy sustains stable system performance under unexpected disruptions across diverse public spaces, showing strong adaptability and reliability.

The trend analysis across the three scenarios shows that the proposed closed-loop air-ground-human collaborative strategy consistently achieves high coverage, rapid response, superior work efficiency, high resource utilization, balanced task allocation, and strong system robustness. Compared with the other three methods, it demonstrates significant advantages in adaptability and stability, highlighting that the case-density-driven closed-loop intelligent disinfection strategy possesses strong scenario generalization capabilities, suitable for complex and dynamic public health environments.

3.4.2 Verification of efficiency and robustness of the closed-loop mechanism

To validate the advantages of the closed-loop intelligent disinfection strategy in terms of work efficiency and system robustness, a comparative experiment was designed between closed-loop and non-closed-loop methods. The non-closed-loop method removes the closed-loop replanning module from the multi-objective scheduling strategy, while the closed-loop method corresponds to the proposed air-ground-human collaborative closed-loop scheduling strategy.

The experimental scenarios remained the three typical environments A, B, and C. Each scenario ran for 120 minutes, with task execution states recorded throughout. The results are summarized in Table 4.

The data indicate that introducing the closed-loop mechanism significantly improves WE and RI across all three scenarios, highlighting the core value of the closed-loop strategy in enhancing operational performance and system resilience.

From the experimental data, it can be observed that the closed-loop intelligent disinfection strategy significantly improves WE and RI across the three typical scenarios, fully demonstrating the core value of the closed-loop

mechanism.

In scenario A, the closed-loop mechanism increased WE from 7.92 ha/h to 10.54 ha/h, an improvement of approximately 33%, while RI rose from 0.59 to 0.88, indicating stronger system adaptability in responding to unexpected events. In scenario B, WE increased from 8.03 ha/h to 10.78 ha/h and RI from 0.61 to 0.87. In scenario C, WE rose from 8.08 ha/h to 10.65 ha/h and RI from 0.60 to 0.87. The comparisons across the three scenarios indicate that the closed-loop mechanism effectively enhances disinfection efficiency and strengthens system stability and continuous operation capability across different environment types.

Table 4. Verification of closed-loop mechanism on performance metrics

Scenario	Method	CR	TTC(min)	WE(ha/h)	RU(%)	TB	RI
A	Without closed-loop	0.77 ± 0.04	124.56 ± 14.32	7.92 ± 0.71	71.20 ± 4.00	0.63 ± 0.05	0.59 ± 0.06
	With closed-loop	0.87 ± 0.03	63.48 ± 10.57	10.54 ± 0.98	83.90 ± 2.85	0.82 ± 0.04	0.88 ± 0.03
B	Without closed-loop	0.78 ± 0.04	119.48 ± 14.72	8.03 ± 0.72	72.13 ± 3.91	0.65 ± 0.05	0.60 ± 0.06
	With closed-loop	0.88 ± 0.03	61.32 ± 10.23	10.78 ± 0.95	84.25 ± 2.91	0.83 ± 0.04	0.87 ± 0.03
C	Without closed-loop	0.78 ± 0.04	121.73 ± 14.89	8.08 ± 0.70	71.88 ± 3.95	0.65 ± 0.05	0.60 ± 0.06
	With closed-loop	0.88 ± 0.03	62.47 ± 10.45	10.65 ± 0.98	83.92 ± 2.87	0.82 ± 0.04	0.87 ± 0.03

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

Further analysis of other performance metrics shows that, alongside improvements in efficiency and robustness, the closed-loop mechanism also drives enhancements in CR, TTC, TB, and RU. Across the three scenarios, CR increased by approximately 0.09–0.10, demonstrating that the closed-loop strategy achieves more comprehensive coverage of high-risk areas. TTC was reduced by nearly half, showing that the mechanism significantly shortens the response time from case detection to task completion. Both TB and RU improved noticeably, indicating that the closed-loop mechanism enables balanced task allocation across multiple platforms while enhancing resource utilization.

Mechanistically, the advantages of the closed-loop strategy arise from three aspects.

Dynamic task adjustment: By real-time monitoring of case distribution, task execution status, and population mobility, the closed-loop mechanism dynamically reallocates tasks, ensuring reasonable division of labor among platforms and maximizing operational efficiency.

Rolling replanning for disturbances: In the event of drone failures, insufficient disinfectant, or temporary area lockdowns, the mechanism can perform rolling replanning to quickly reschedule affected tasks, significantly enhancing system robustness.

Multi-objective coupled optimization: By simultaneously considering coverage, efficiency, resource utilization, and task balance, the mechanism ensures continuity and reliability of the disinfection system under high-risk conditions.

Overall, the closed-loop mechanism not only shows significant advantages in WE and RI but also achieves multi-dimensional improvements in efficiency, robustness, and operational quality through coordinated optimization of CR, TTC, TB, and RU. The data clearly demonstrate that air-ground-human closed-loop collaborative scheduling is a key technological approach for enhancing the intelligence and effectiveness of public health disinfection operations.

3.4.3 Parameter sensitivity analysis

To further evaluate the stability and robustness of the closed-loop intelligent disinfection strategy, a sensitivity analysis was conducted on key parameters of the strategy. The focus was on examining the effects of risk field rolling update frequency, task allocation weighting coefficients, and rolling replanning time intervals on system performance. Experiments were conducted in the three typical scenarios (A, B, C), with each scenario running for 120 minutes. Six performance metrics were recorded: CR, TTC, WE, RU, TB, and RI. The experimental results are presented in Table 5 and Figure 2.

As the update frequency was shortened from 30 minutes to 5 minutes, the system responded more promptly to changes in case distribution and population movement. The table data show that WE increased from 10.05 ha/h to 10.62 ha/h, CR rose from 0.85 to 0.88, and RI improved from 0.86 to 0.88. TB and RU also experienced slight improvements. Overall, higher update frequencies led to better performance in work efficiency and system robustness, though excessively high frequencies could increase computational overhead, so practical deployment requires a balance between performance and computational load. When the update frequency is high, the CR, TB, and RI curves in the left figure remain at high levels, indicating that the closed-loop strategy can promptly perceive changes in cases and population flow, rapidly adjust tasks, improve coverage and task balance, and enhance system robustness. The right figure shows that under high update frequency, WE is clearly higher than in low-frequency scenarios, TTC is shorter, and RU remains stable, indicating improved work efficiency and resource utilization. As

the update frequency decreases, all metric curves slightly decline due to delayed responses, but overall performance remains at a relatively high level.

Adjusting the weights of work efficiency, coverage, task balance, and resource utilization by $\pm 20\%$ shows that WE is most sensitive to weight changes, varying from 10.21 ha/h to 10.69 ha/h, while CR and TB fluctuate slightly, and RI remains high at 0.87–0.88. This demonstrates that the closed-loop scheduling maintains strong robustness under multi-objective constraints, and system performance remains stable even with weight variations. Adjusting task allocation weights has a relatively mild effect on the metrics. In the left figure, CR, TB, and RI curves show slight fluctuations, indicating that the closed-loop strategy maintains high robustness and task balance despite weight adjustments. In the right figure, WE is most sensitive to weight changes, slightly increasing under higher weight configurations, while TTC decreases and RU remains largely unchanged. Thus, weighting coefficients primarily optimize work efficiency and resource scheduling, with limited effect on overall system stability.

Table 5. Sensitivity analysis of key parameters on performance metrics

Parameter	Value/Setting	CR	TTC(min)	WE(ha/h)	RU(%)	TB	RI
Risk field update frequency	5 min	0.88 ± 0.03	62.15 ± 28.92	10.62 ± 0.98	84.03 ± 0.04	0.82 ± 0.04	0.88 ± 0.03
	10 min	0.87 ± 0.03	63.52 ± 10.63	10.51 ± 0.91	83.73 ± 0.04	0.81 ± 0.04	0.88 ± 0.03
	15 min	0.86 ± 0.04	65.18 ± 10.32	10.82 ± 0.88	82.98 ± 0.80	0.87 ± 0.05	0.87 ± 0.03
	30 min	0.85 ± 0.04	67.95 ± 11.48	10.25 ± 0.91	82.03 ± 0.79	0.86 ± 0.06	0.86 ± 0.04
Task allocation weight	Lower	0.86 ± 0.04	63.97 ± 10.81	10.21 ± 0.85	82.98 ± 0.98	0.79 ± 0.81	0.88 ± 0.03
	Nominal	0.87 ± 0.03	63.48 ± 10.60	10.53 ± 0.90	83.68 ± 0.91	0.81 ± 0.04	0.88 ± 0.03
	Higher	0.88 ± 0.03	62.83 ± 10.69	10.89 ± 0.89	84.18 ± 0.04	0.82 ± 0.04	0.88 ± 0.03
Rolling replanning interval	2 min	0.88 ± 0.03	61.05 ± 10.61	10.61 ± 0.81	83.52 ± 0.82	0.82 ± 0.08	0.88 ± 0.03
	5 min	0.88 ± 0.03	62.08 ± 10.33	10.59 ± 0.88	84.02 ± 0.82	0.82 ± 0.08	0.88 ± 0.03
	10 min	0.87 ± 0.03	63.49 ± 10.50	10.50 ± 0.90	83.70 ± 0.81	0.81 ± 0.04	0.88 ± 0.03
	20 min	0.85 ± 0.04	67.03 ± 11.23	10.12 ± 0.92	82.51 ± 0.80	0.86 ± 0.05	0.86 ± 0.03
	30 min	0.84 ± 0.04	70.02 ± 11.51	9.91 ± 0.90	81.97 ± 0.79	0.85 ± 0.05	0.85 ± 0.04

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

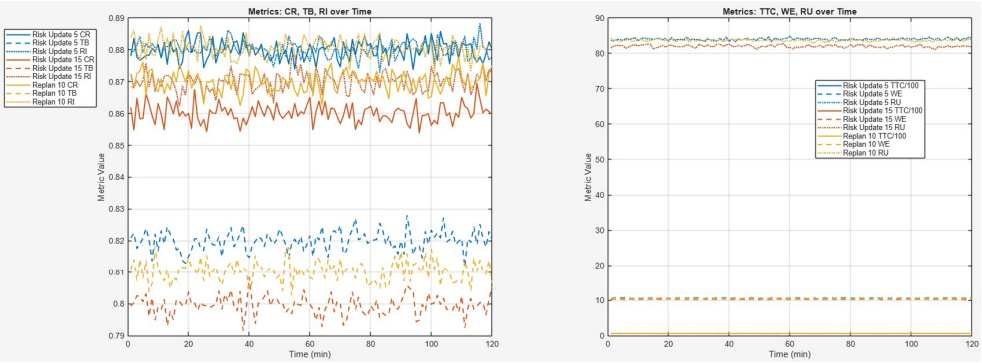


Figure 2. Effect of key parameters on efficiency and robustness over time in the closed-loop intelligent disinfection strategy

The rolling replanning interval significantly impacts the response speed of the closed-loop strategy. Shortening the interval from 30 minutes to 2 minutes increases WE from 9.91 ha/h to 10.61 ha/h, RI from 0.85 to 0.88, and CR also rises while TTC decreases. Short intervals allow rapid task adjustments to handle emergencies, though computational overhead rises; overly long intervals reduce work efficiency and system robustness. Overall, a replanning interval of around 10 minutes strikes a good balance between performance and computational load. The length of the replanning interval significantly affects the system's response to unexpected events. Short intervals keep CR, TB, and RI curves high (left figure), WE peaks higher, TTC lower (right figure), and RU stable, indicating rapid task adjustment, improved efficiency, and sustained robustness. Longer intervals cause slight drops in curves, especially in WE and TTC, though the system still maintains a certain level of robustness.

From the sensitivity analysis, the closed-loop intelligent disinfection strategy demonstrates good stability and robustness under variations of key parameters. Risk field update frequency and rolling replanning interval are the main factors affecting work efficiency and system robustness, while task allocation weighting coefficients serve to balance multi-objective goals. The closed-loop strategy shows strong robustness and stability across parameter

changes. Risk field update frequency and rolling replanning interval mainly influence efficiency metrics (WE, TTC, RU) as task allocation weights primarily regulate work efficiency. Across all scenarios, the closed-loop mechanism ensures high levels of coverage, task balance, and system robustness, providing reliable support for practical deployment.

3.4.4 Ablation study

To quantify the contribution of each key module in the closed-loop intelligent disinfection strategy to overall system performance, an ablation study was designed. The full system was compared against versions with individual modules removed. Experiments were conducted in three typical scenarios: A (high-density urban area), B (mixed old communities and market areas), and C (large-scale public activity areas). Each scenario ran for 120 minutes, and six core performance metrics were recorded: CR, TTC, WE, RU, TB, and RI.

Experimental Design:

Full System: complete system including rolling risk field updates, closed-loop replanning, and task demand prediction modules.

Without Risk Update: rolling risk field update module removed, to evaluate its contribution to system performance.

Without Replanning: closed-loop replanning module removed, to assess its effect on efficiency and robustness.

Without Prediction: task demand prediction module removed, to measure its impact on task balance and work efficiency.

The ablation experiment results for the three scenarios are presented in Table 6, 7, and 8.

Table 6. Ablation study results—scenario A (dense urban area)

Method	CR	TTC (min)	WE (ha/h)	RU (%)	TB	RI
Full system	0.87 ± 0.03	63.48 ± 10.57	10.54 ± 0.92	83.67 ± 2.85	0.81 ± 0.04	0.88 ± 0.03
Without risk update	0.84 ± 0.04	72.15 ± 12.23	9.83 ± 0.95	81.42 ± 3.12	0.77 ± 0.05	0.83 ± 0.04
Without replanning	0.82 ± 0.04	85.32 ± 14.18	9.12 ± 0.88	79.05 ± 3.45	0.74 ± 0.05	0.80 ± 0.05
Without prediction	0.85 ± 0.03	68.47 ± 11.02	10.01 ± 0.90	82.11 ± 2.97	0.79 ± 0.04	0.85 ± 0.04

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

Table 7. Ablation study results—scenario B (mixed old community & market area)

Method	CR	TTC (min)	WE (ha/h)	RU (%)	TB	RI
Full system	0.88 ± 0.03	61.32 ± 10.23	10.78 ± 0.95	84.25 ± 2.91	0.83 ± 0.04	0.87 ± 0.03
Without risk update	0.85 ± 0.04	69.87 ± 12.18	10.05 ± 0.92	81.95 ± 3.18	0.79 ± 0.05	0.83 ± 0.04
Without replanning	0.83 ± 0.04	82.54 ± 13.95	9.30 ± 0.90	78.82 ± 3.40	0.76 ± 0.05	0.80 ± 0.05
Without prediction	0.86 ± 0.03	66.21 ± 11.10	10.20 ± 0.93	82.55 ± 2.95	0.80 ± 0.04	0.85 ± 0.04

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

Table 8. Ablation study results—scenario C (large public activity area)

Method	CR	TTC (min)	WE (ha/h)	RU (%)	TB	RI
Full system	0.88 ± 0.03	62.47 ± 10.45	10.65 ± 0.98	83.92 ± 2.87	0.82 ± 0.04	0.87 ± 0.03
Without risk update	0.85 ± 0.04	71.23 ± 12.35	9.92 ± 0.95	81.55 ± 3.20	0.78 ± 0.05	0.83 ± 0.04
Without replanning	0.83 ± 0.04	83.47 ± 13.88	9.25 ± 0.91	78.95 ± 3.38	0.75 ± 0.05	0.80 ± 0.05
Without prediction	0.86 ± 0.03	65.32 ± 11.15	10.15 ± 0.94	82.20 ± 2.92	0.80 ± 0.04	0.85 ± 0.04

Note: CR = Coverage rate, TTC = Time-to-control, WE = Work efficiency, RU = Resource utilization, TB = Task balance, and RI = Robustness index

The experimental results indicate that the full system achieved the best performance across all scenarios, with CR reaching 0.87–0.88, TTC maintained at 61–63 minutes, WE around 10.5–10.8 ha/h, RU approximately 83–84%, TB about 0.81–0.83, and RI around 0.87–0.88. This demonstrates that the coordinated operation of all modules enables high coverage, rapid suppression, efficient execution, and system stability.

When the rolling risk field update module was removed, both CR and TB decreased, indicating that real-time risk perception plays a critical role in task allocation accuracy and balance. Meanwhile, WE and RI also declined to some extent, showing that the risk update module is key to improving work efficiency and system robustness. Removing the closed-loop replanning module had the most significant impact: TTC increased noticeably, work efficiency dropped,

and RI decreased, highlighting that the replanning module is essential for rapid response to unexpected events and maintaining system stability. In contrast, removing the task demand prediction module caused slight decreases in CR and TB, with minimal impact on WE and RI, indicating that the prediction module primarily enhances the foresight and balance of task allocation, with limited effect on core control and overall robustness.

Across the three scenarios, the experimental results reveal that each module contributes differently to the closed-loop intelligent disinfection strategy.

The rolling risk field update module significantly improves coverage, task balance, and system robustness, effectively enhancing the overall stability and reliability of the system.

Closed-loop replanning module has the greatest impact on suppression timeliness, work efficiency, and system robustness, serving as the core for high-efficiency and stable operation.

Task demand prediction module assists in enhancing task balance and work efficiency, but its influence on overall system stability is limited.

Overall, the ablation experiments confirm that the closedloop air-ground-human collaborative strategy maintains high efficiency and robustness in complex urban environments, while clearly delineating the specific contribution of each module to overall performance.

To provide a more intuitive comparison of each module's contribution to overall performance across different scenarios, the results of the full system and ablation variants were visualized in a radar chart. The six axes correspond to CR, TTC, WE, RU, TB, and RI, with the area and shape of the radar chart illustrating each module's performance across the metrics, facilitating analysis of their combined impact on efficiency and robustness, as shown in Figure 3.

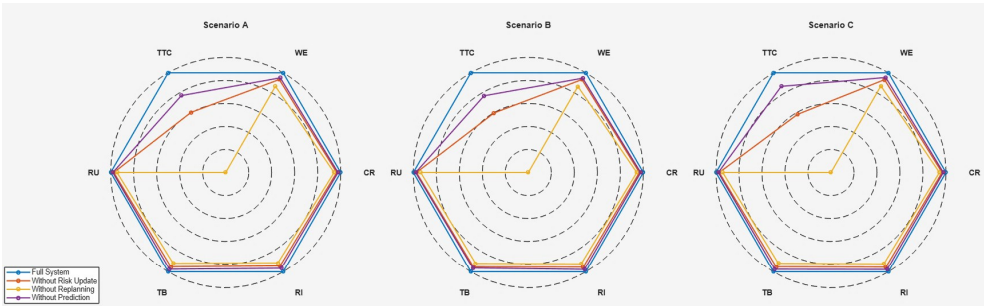


Figure 3. Ablation study of closed-loop air-ground-human collaborative disinfection strategy

The radar chart displays the performance of six key metrics (CR, WE, TTC, RU, TB, RI). The full system consistently performs best across all scenarios, demonstrating clear advantages in coverage, work efficiency, suppression timeliness, resource utilization, task balance, and robustness. This highlights the critical role of the closed-loop mechanism and multi-module coordination in overall disinfection effectiveness.

Removing the rolling risk field update module reduces coverage and task balance, indicating that real-time perception of case distribution and population flow is crucial for prioritizing high-risk areas. Without the closed-loop replanning module, TTC increases sharply and system robustness declines, showing that replanning dynamically adjusts task schedules to respond to unexpected events and ensure continuous operation. Removing the prediction module slightly reduces work efficiency, but coverage and robustness remain close to the full system, indicating that prediction enables proactive task planning and improves multi-agent coordination efficiency.

Across scenarios, including dense urban areas, mixed old communities with markets, and large public activity areas, the ablation trends are consistent, verifying the robustness of the closed-loop strategy under different environments. In summary, the radar chart intuitively illustrates the contributions of the closed-loop mechanism and each key module to the air-groundhuman collaborative disinfection strategy, emphasizing the core value of risk perception, rolling replanning, and task prediction in enhancing operational efficiency and system robustness.

4 Discussion and Conclusions

This study proposed a case-density-driven, closed-loop air-ground-human collaborative disinfection strategy, establishing an end-to-end framework that integrates case perception, risk field modeling, task allocation, and execution monitoring. The framework enables dynamic optimization and intelligent scheduling of public health disinfection tasks. Simulation experiments across three representative scenarios-dense urban areas (scenario A), mixed old communities and market regions (scenario B), and large public activity areas (scenario C)demonstrated that the closed-loop strategy outperforms baseline methods in key performance metrics, including WE, CR, RU, and system RI.

The observed performance improvements can be attributed to three main mechanisms.

Rolling risk field updates allow the system to promptly identify high-risk areas and prioritize resource allocation, ensuring timely responses to dynamic case distributions.

Rolling replanning ensures task continuity and feasibility under unexpected events or resource fluctuations, such as sudden area lockdowns, equipment failures, or limited disinfectant availability.

Task demand prediction enhances the foresight of task allocation, improving both work efficiency and multi-agent coordination.

The study's innovations can be summarized as follows.

End-to-end closed-loop design: by tightly coupling case data perception, risk modeling, multi-agent task allocation, rolling replanning, and execution monitoring, the framework establishes a complete closed-loop control from data acquisition to task completion. This design not only improves coverage and efficiency but also strengthens system robustness in dynamic and uncertain environments.

Multi-agent collaborative optimization: the proposed multiobjective model considers air drones, ground vehicles, and human teams, integrating efficiency, coverage continuity, task balance, resource utilization, and robustness. Experimental results show that this approach effectively allocates tasks while balancing efficiency and fairness across heterogeneous platforms.

Dynamic rolling replanning: the strategy adapts task schedules based on real-time case information, population movement, and execution states. This adaptability ensures effective responses to unforeseen events and enhances operational efficiency.

Despite these contributions, several limitations remain. Current experiments rely on simulated population mobility and case distributions, which may not fully capture the complexity of real-world urban environments. Additionally, the strategy has yet to be validated in operational public health settings, and the integration with predictive epidemiological models has not been fully explored.

This work presents a closed-loop air-ground-human collaborative disinfection strategy driven by case density, demonstrating significant improvements in coverage, efficiency, and robustness across diverse urban scenarios. The proposed framework integrates real-time risk perception, multi-agent task allocation, rolling replanning, and predictive task scheduling to achieve high adaptability in dynamic and uncertain environments.

Key conclusions are as follows.

Module synergy is critical: rolling risk field updates, closedloop replanning, and task prediction each contribute differently to overall performance, with replanning playing a core role in maintaining robustness and efficiency.

Closed-loop strategy ensures stability: across dense urban, mixed, and large public areas, the framework consistently maintains high coverage, rapid response, balanced task distribution, and robust operation under simulated disturbances.

Practical applicability: the framework demonstrates potential for real-world deployment in public health management, offering an adaptable and efficient solution for multi-agent collaborative disinfection tasks.

Future work will focus on the following aspects.

Integration with real-world data: incorporating actual urban population mobility and case data to validate scalability and stability in large-scale environments.

Field trials with public health agencies: assessing operational feasibility, efficiency, and decision-making support in real-world scenarios.

Predictive and proactive scheduling: leveraging epidemiological forecasts to anticipate high-risk areas and optimize task allocation proactively.

Further enhancements: exploring multi-strategy fusion, energy-constrained optimization, and cross-region collaborative scheduling to improve overall system performance and adaptability.

Overall, this study provides a comprehensive framework for intelligent public health disinfection, highlighting the significance of closed-loop multi-agent coordination in enhancing efficiency, robustness, and situational adaptability.

Author Contributions

Conceptualization, L.H.Z.; methodology, N.F.Z.; software, J.F.Y.; validation, Z.Q.L.; data curation, X.L., L.H.Z., and Y.Y.W.; writing original draft preparation, L.H.Z.; writing review and editing, Z.Q.L. and Y.Y.W.; visualization, N.F.Z. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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